

Hybrid ISI Cancellation and Reservoir Computing for Mobile Molecular Communication Channels with Smart Transmitter Extension

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Abstract—Molecular communication via diffusion (MCvD) enables information transfer through chemical signaling between nanoscale devices, offering a biocompatible alternative to electromagnetic communication in environments such as the human bloodstream. The primary challenge in MCvD is inter-symbol interference (ISI), caused by residual molecules lingering in the channel from previous transmissions. While mathematical ISI cancellation techniques exist for static channels, they fail under mobile conditions where the transmitter–receiver distance varies stochastically. This paper proposes a hybrid detection architecture that combines the ISI cancellation preprocessing filter of Meng et al. (2014) with the Reservoir Computing (RC) detector of Bilge et al. (2025). We further introduce a *Smart Transmitter* mechanism that proactively adjusts emission intensity based on estimated residual ISI, shifting intelligence from the receiver to the transmitter side. Through a comprehensive benchmark of ten detection methods across varying ISI severities, we demonstrate that the hybrid detector achieves up to 100% accuracy at moderate ISI with only 401 trainable parameters, and that the Smart Transmitter provides additional gains—particularly for sequential-memory detectors—by reducing the dynamic range the receiver must handle. These preliminary results suggest that combining physics-informed preprocessing with lightweight machine learning offers a practical path toward reliable, low-complexity nanoscale communication.

Index Terms—Molecular communication, inter-symbol interference, reservoir computing, echo state network, ISI cancellation, smart transmitter, nanonetworks

I. INTRODUCTION

Molecular Communication via Diffusion (MCvD) is a bio-inspired paradigm in which information is encoded in the release of chemical molecules by a transmitter (TX) and decoded by a receiver (RX) that counts absorbed molecules [1], [2]. This approach is uniquely suited to nanoscale networks operating in biological environments—such as targeted drug delivery, intra-body health monitoring, and lab-on-a-chip systems—where conventional electromagnetic signaling is impractical due to energy, size, and biocompatibility constraints.

The fundamental challenge in MCvD is *Inter-Symbol Interference* (ISI). When the transmitter releases molecules to encode a binary “1,” the molecules propagate through Brownian diffusion and do not instantaneously disappear after reception. Instead, a heavy-tailed residual concentration persists in the channel, contaminating subsequent symbol slots. As the symbol rate increases (i.e., the bit interval T_b decreases),

this chemical echo grows progressively more severe, driving detection accuracy toward random guessing.

Prior work has addressed ISI from two distinct angles:

- **Mathematical ISI Cancellation** (Meng et al., 2014 [1]): A lightweight technique that subtracts the estimated ISI baseline from the received signal. Effective for *static* channels where the channel impulse response (CIR) is time-invariant, but degrades under mobility.
- **Data-Driven Detection via Reservoir Computing** (Bilge et al., 2025 [2]): An Echo State Network (ESN) approach that learns the time-varying channel dynamics with only 401 trainable parameters—dramatically fewer than deep learning alternatives (e.g., MLP with 264,000 parameters)—while matching or exceeding their accuracy.

Our Contributions. This paper makes the following contributions:

- 1) We propose a *Hybrid ISI-Cancel + RC* detection architecture that applies the Meng 2014 ISI cancellation as a preprocessing filter before feeding the cleaned signal into the Bilge 2025 RC-ISI detector. The hypothesis is that physics-informed preprocessing reduces the dynamic range the reservoir must cover, improving accuracy without increasing model complexity.
- 2) We introduce a *Smart Transmitter* mechanism in which the transmitter proactively adjusts the number of emitted molecules based on its estimate of residual ISI from past transmissions. This shifts computational burden from the resource-constrained receiver to a potentially more capable transmitter.
- 3) We present a comprehensive benchmark of ten detection methods—including classical threshold, deep learning, and reservoir computing variants—under both normal and Smart TX operation, across a range of ISI severities.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the system model and channel simulation. Section IV presents the detection methods. Section V details the proposed hybrid architecture. Section VI introduces the Smart Transmitter. Section VII provides preliminary results. Section VIII discusses the findings. Section IX concludes the paper.

II. RELATED WORK

A. ISI Cancellation for Static Channels

Meng et al. [1] proposed ISI mitigation techniques for 3D diffusion-based molecular communication with a static point transmitter and a spherical absorbing receiver. Their ISI cancellation scheme exploits the fact that, in a static channel, the residual concentration from previous symbols evolves predictably. By sampling the signal at the beginning of each symbol slot (capturing the ISI baseline) and subtracting it from the peak observation, the receiver effectively isolates the contribution of the current bit. This simple difference operation,

$$z[k] = u[k] - u[k-1], \quad (1)$$

where $u[k]$ is the end-of-symbol received concentration for the k -th bit, removes the slowly-varying ISI tail with zero additional parameters.

The limitation is that Equation (1) assumes a time-invariant CIR. When TX and RX are mobile, the CIR changes at every symbol, and the subtraction operates on mismatched channel states, leaving significant residual interference.

B. Reservoir Computing for Mobile Channels

Bilge et al. [2] addressed mobile molecular communication channels using Reservoir Computing (RC), specifically a Leaky-Integrator Echo State Network (LI-ESN). The key architectural insight is that only the linear readout layer is trained via Ridge regression (closed-form solution), while the recurrent reservoir weights remain fixed. This eliminates backpropagation entirely, yielding:

- Ultra-fast training (Ridge regression, no iterative optimization);
- Extremely low model complexity (401 parameters: 400 readout weights + 1 bias);
- Temporal memory through recurrent reservoir dynamics.

The RC-ISI variant further optimizes the decision threshold using ROC analysis on a validation set, minimizing BER rather than using a fixed 0.5 threshold. The authors demonstrated that RC-ISI outperforms MLP, LSTM, and CNN detectors on Smoldyn-simulated mobile channels, particularly under severe ISI conditions ($T_b = 10$ s), while requiring $600\times$ fewer parameters than MLP.

C. Simulation Environments

The standard simulation tool for MCvD research is Smoldyn [3], a particle-based stochastic simulator that tracks individual molecules in 3D space. While physically accurate, Smoldyn simulations are computationally expensive (hours per experiment). An alternative is the MUCIN (MolecUlar CommunicatIoN) Simulator for MATLAB [4], which provides end-to-end MCvD simulation with first-hitting processes and receptor kinetics. In this work, we employ an analytically-calibrated Python channel model that captures the essential physics—ISI memory, TX/RX mobility, and noise—while enabling rapid experimentation (0.03s per 2000 symbols vs. hours for Smoldyn).

III. SYSTEM MODEL

A. Channel Model

We consider a diffusion-based MCvD system within a bounded cubic domain of half-side $L = 100\mu\text{m}$, consistent with the Smoldyn environment in [2]. The transmitter releases $N = 2000$ molecules to encode bit “1” and zero molecules for bit “0.” The receiver is a fully absorbing sphere of radius $r_0 = 5\mu\text{m}$ centered at the origin.

1) *ISI Memory Model*: The effective memory of the bounded channel is governed by the domain diffusion relaxation time:

$$\tau = \frac{L^2}{D_M}, \quad (2)$$

where $D_M = 1.01 \times 10^{-9} \text{m}^2/\text{s}$ is the molecular diffusion coefficient. For the default parameters, $\tau \approx 9.9$ s. The per-symbol ISI decay factor at TX–RX distance r is modeled as:

$$\rho(r) = \exp\left(-\frac{T_b}{\tau_{\text{eff}}(r)}\right), \quad \tau_{\text{eff}}(r) = \tau \cdot \left(\frac{r}{r_{\text{ref}}}\right)^2, \quad (3)$$

where $r_{\text{ref}} = 20\mu\text{m}$ is the reference distance. This yields $\rho \approx 0.37$ at $T_b = 10$ s (severe ISI) and $\rho \approx 4.5 \times 10^{-5}$ at $T_b = 100$ s (negligible ISI).

2) *Signal Model*: The received signal at symbol k is:

$$s[k] = \frac{\sum_{j=0}^k b[j] \cdot \rho_k^{(k-j)}}{\sum_{m=0}^k \rho_k^m} + n[k], \quad (4)$$

where $b[j] \in \{0, 1\}$ is the j -th transmitted bit, $\rho_k = \rho(r[k])$ is the decay factor at the instantaneous distance, and $n[k] \sim \mathcal{N}(0, \sigma_n^2)$ is additive Gaussian noise with $\sigma_n = 0.03$. The denominator normalizes the signal to $[0, 1]$.

3) *Mobility Model*: Both TX and RX undergo independent Brownian motion at the symbol rate. The TX–RX distance $r[k]$ follows a random walk:

$$r[k] = r[k-1] + \Delta r[k], \quad \Delta r \sim \mathcal{N}(0, \sigma_{\text{walk}}^2), \quad (5)$$

where $\sigma_{\text{walk}} = \sqrt{2(D_T + D_R)T_b}$, with $D_T = 4.74 \times 10^{-14} \text{m}^2/\text{s}$ and $D_R = 2.31 \times 10^{-12} \text{m}^2/\text{s}$ being the TX and RX diffusion coefficients, respectively. The distance is clamped to $[2r_0, 1.5L]$ to maintain physical plausibility.

This mobility model introduces a time-varying CIR: at $T_b = 100$ s, $\sigma_{\text{walk}} \approx 21.7\mu\text{m} \approx r_{\text{ref}}$, meaning the channel can change dramatically between consecutive symbols.

IV. DETECTION METHODS

We benchmark ten detection methods spanning classical, deep learning, and reservoir computing paradigms.

A. Classical Detectors

Peak-Fixed (Threshold). The simplest detector applies a fixed threshold θ to the received signal: $\hat{b}[k] = \mathbb{1}\{s[k] \geq \theta\}$. The threshold is optimized via grid search on training data.

Adaptive-EMA (Damrath & Hoeher, 2016). An exponential moving average tracks the ISI baseline $I[k] = \beta I[k-1] + (1 - \beta)s[k-1]$, and the decision rule is $\hat{b}[k] = \mathbb{1}\{s[k] > \eta + I[k]\}$. Both β and η are tuned via grid search.

MAP (Mismatched). A Viterbi-style detector that estimates a static average CIR from training data via least-squares regression and performs symbol-by-symbol MAP decisions using a Gaussian likelihood model. Optimal for static channels but severely mismatched under mobility.

B. Deep Learning Detectors

Feedforward-MLP. A three-layer fully connected network (input $\rightarrow 128 \rightarrow 64 \rightarrow 1$) with ReLU activations and sigmoid output, operating on sliding windows of past observations.

ANN. A shallower variant (input $\rightarrow 32 \rightarrow 16 \rightarrow 1$) with the same window-based architecture.

CNN. Two 1-D convolutional layers (8 filters, kernel size 3) followed by a linear readout, capturing local temporal patterns.

LSTM. A recurrent network with 16 hidden units processing sequences of length 10, capable of modeling long-range temporal dependencies.

All deep learning models are trained with Adam (lr = 10^{-3} , 30 epochs, batch size 64) and BCE loss.

C. Reservoir Computing Detectors

RC. The standard Echo State Network detector with a fixed reservoir ($N_r = 400$ nodes, spectral radius $\rho_{sr} = 0.7$, leaky rate $\alpha = 0.3$, sparsity 0.1). Only the linear readout is trained via Ridge regression ($\lambda = 10^{-4}$). Decision threshold fixed at 0.5.

RC-ISI. Identical to RC but with ROC-optimized threshold: a validation split is used to find the threshold η^* minimizing $\text{BER} = \frac{1}{2}(\text{FPR} + \text{FNR})$.

V. PROPOSED HYBRID ARCHITECTURE

Our primary contribution is the *Hybrid ISI-Cancel + RC* detector, which combines the mathematical rigor of Meng 2014 with the adaptive learning capability of Bilge 2025. The detection pipeline is:

$$\begin{aligned} u[k] &\xrightarrow{\text{ISI Cancel}} z[k] = u[k] - u[k-1] \\ &\xrightarrow{\text{Z-score}} \tilde{z}[k] \xrightarrow{\text{ESN}} \mathbf{x}[k] \\ &\xrightarrow{\text{Ridge}} \hat{y}[k] \xrightarrow{\eta^*} \hat{b}[k] \end{aligned} \quad (6)$$

A. Design Rationale

The ISI cancellation step in Equation (1) was originally designed for static channels. In a mobile channel, the subtraction $z[k] = u[k] - u[k-1]$ does not perfectly cancel ISI because the CIR varies between symbols. However, it still achieves two useful effects:

- 1) **Baseline removal:** The slowly-varying ISI accumulation is partially suppressed, emphasizing rapid transitions (i.e., new bit arrivals).
- 2) **Dynamic range reduction:** The cleaned signal $z[k]$ has a narrower amplitude range, reducing the burden on the downstream reservoir.

The key insight is the *division of labor*: the ISI cancellation handles the deterministic, slowly-varying component of ISI, while the reservoir models the residual stochastic, time-varying interference that the simple subtraction cannot capture.

B. Complexity Analysis

The hybrid detector adds *zero* trainable parameters compared to RC-ISI. The ISI cancellation step is a single subtraction ($O(1)$ per symbol), and the Z-score normalization is a fixed affine transform. The total parameter count remains 401 (400 readout weights + 1 bias), identical to RC and RC-ISI.

VI. SMART TRANSMITTER

Traditional MCvD system design places all intelligence at the receiver. However, the transmitter possesses information that the receiver lacks: knowledge of its own past transmissions. We propose exploiting this asymmetry through an *ISI-Aware Smart Transmitter* that adjusts emission intensity proactively.

A. Emission Control Mechanism

In the standard model, the transmitter emits a fixed number $N_{\text{base}} = 2000$ molecules for every bit “1.” In the Smart TX model, the transmitter estimates the residual ISI from its own transmission history and reduces emission accordingly:

$$N_{\text{emit}}[k] = N_{\text{base}} \cdot \max\left(\omega_{\min}, 1 - \gamma \cdot \hat{C}_{\text{ISI}}[k]\right), \quad (7)$$

where:

- $\hat{C}_{\text{ISI}}[k]$ is the transmitter’s estimate of the normalized ISI baseline at symbol k , computed from its own past bit decisions and emission weights using the reference decay factor $\rho_{\text{ref}} = \rho(r_{\text{ref}})$:

$$\hat{C}_{\text{ISI}}[k] = \frac{\sum_{j=0}^{k-1} b[j] \cdot w[j] \cdot \rho_{\text{ref}}^{(k-j)}}{\sum_{m=0}^{k-1} \rho_{\text{ref}}^m}, \quad (8)$$

where $w[j] \in [\omega_{\min}, 1]$ is the emission weight used for bit j ;

- $\gamma = 0.8$ controls the aggressiveness of the adjustment;
- $\omega_{\min} = 0.1$ ensures a minimum emission of 10% of N_{base} .

B. Key Design Choices

Transmitter-side only: The transmitter uses only information available to it—its own bit sequence and a reference channel model (ρ_{ref}). It does *not* require feedback from the receiver, preserving the simplicity of the communication protocol.

No receiver modification: All existing detectors operate unchanged. The Smart TX modifies only the channel input, making it compatible with any detection scheme.

Proactive vs. reactive: Unlike receiver-side ISI cancellation (reactive), Smart TX preemptively reduces the ISI contribution before molecules enter the channel (proactive). These approaches are complementary and can be combined.

VII. PRELIMINARY RESULTS

We present preliminary results from a fast benchmark (500 bits, random seed 42, 70/30 train/test split) at three representative symbol intervals: $T_b \in \{10, 100, 200\}$ s, corresponding to severe, moderate, and low ISI conditions, respectively.

TABLE I: Detection Accuracy (%) — Normal TX vs. Smart TX

	$T_b = 10$ s		$T_b = 100$ s		$T_b = 200$ s	
	N	S	N	S	N	S
Peak-Fixed	60.0	59.3	92.0	92.0	92.7	92.7
Adapt.-EMA	76.7	76.0	96.0	96.0	99.3	99.3
MAP (Mism.)	58.0	62.0	92.7	92.7	96.0	96.0
FF-MLP	68.7	68.0	84.7	85.3	88.7	88.0
ANN	62.0	60.7	84.0	84.7	86.7	90.0
CNN	70.0	72.0	82.0	77.3	84.0	78.7
LSTM	68.7	70.7	88.7	92.7	94.0	91.3
RC	78.7	78.7	97.3	97.3	98.7	98.7
RC-ISI	74.7	78.7	97.3	97.3	98.7	98.7
Hybrid	74.0	74.7	98.7	98.7	100	100

N = Normal TX, S = Smart TX.

A. Detection Accuracy Comparison

Table I summarizes detection accuracy for all ten methods under both Normal and Smart TX operation.

B. Key Observations

1) *Hybrid Detector Performance:* The Hybrid ISI-Cancel+RC detector achieves the highest accuracy at moderate and low ISI ($T_b \geq 100$ s), reaching 98.7% at $T_b = 100$ s and a perfect 100% at $T_b = 200$ s—outperforming all other detectors including RC-ISI (98.7%) at these operating points. Crucially, this improvement comes at *zero additional parameters*: both RC-ISI and Hybrid use exactly 401 parameters.

At severe ISI ($T_b = 10$ s), the hybrid detector does not outperform RC. This is expected: when the channel is highly dynamic (large σ_{walk}), the difference $z[k] = u[k] - u[k-1]$ operates on mismatched CIR states, adding noise rather than removing it. The reservoir alone handles this regime better.

2) *Smart Transmitter Impact:* The Smart TX produces heterogeneous effects across detector families:

Positive gains: RC-ISI improves by +4.0 percentage points (pp) at $T_b = 10$ s (74.7% $\rightarrow$ 78.7%), matching the performance of the standard RC detector. MAP (Mismatched) also gains +4.0 pp at $T_b = 10$ s, consistent with it being most vulnerable to ISI accumulation. LSTM gains +4.0 pp at $T_b = 100$ s.

Neutral: RC and Hybrid detectors show no change—they already achieve near-optimal performance.

Negative: CNN degrades by -4.7 pp at $T_b = 100$ s and -5.3 pp at $T_b = 200$ s. This is attributable to Smart TX altering the signal amplitude distribution: bits with identical values now produce different signal levels, confusing fixed-window convolutional detectors.

3) *Model Complexity:* Table II highlights the efficiency advantage of reservoir-based detectors.

The RC family achieves the best accuracy–complexity trade-off: 401 parameters deliver $\geq 97.3\%$ accuracy at $T_b = 100$ s, while MLP requires 264,449 parameters (659 $\times$ more) for only 84.7%.

TABLE II: Model Complexity and Inference Latency

Detector	Parameters	Latency ($\mu\text{s}/\text{sym}$)
Peak-Fixed / EMA / MAP	0	< 2
RC / RC-ISI / Hybrid	401	~ 16
LSTM	1,233	~ 6
CNN ($T_b = 100$)	8,233	~ 142
MLP ($T_b = 200$)	264,449	~ 3

C. Division of Labor Analysis

The complementary nature of ISI cancellation and Reservoir Computing can be characterized by ISI severity:

- **Low ISI** ($T_b = 200$ s): ISI cancellation effectively removes the residual tail; reservoir fine-tunes the decision. Result: 100% accuracy.
- **Moderate ISI** ($T_b = 100$ s): ISI cancellation partially cleans the signal; reservoir handles the residual time-varying component. Result: 98.7%, vs. 97.3% for RC-ISI alone.
- **Severe ISI** ($T_b = 10$ s): Channel is too dynamic for the static subtraction to help. The reservoir alone is preferable. Result: Hybrid $\approx$ RC-ISI.

This behavior validates the design hypothesis: the hybrid architecture provides the greatest benefit precisely where the channel is stable enough for the ISI cancellation filter to operate meaningfully, but still dynamic enough that a simple threshold fails.

VIII. DISCUSSION

A. Comparison with Literature

Our analytically-calibrated channel produces qualitatively consistent trends with the Smoldyn-based results of Bilge et al. [2]: (i) all detectors degrade as T_b decreases; (ii) RC outperforms deep learning at severe ISI; (iii) MAP (Mismatched) fails catastrophically under mobility. Absolute accuracy values are higher in our simulation because the analytical channel introduces less stochastic noise than particle-level simulation and uses fewer bits (500 vs. 1000 in the paper). The relative ranking of detectors, however, is preserved.

B. Practical Implications

The Smart TX concept has direct implications for asymmetric nanonetwork architectures. In a scenario where the transmitter is an external gateway with ample computational resources and the receiver is a nanoscale in-body device with extreme hardware limitations, Smart TX allows the use of simple, low-power detectors (e.g., Peak-Fixed or Adaptive-EMA) at the receiver while the transmitter handles ISI management. This “smart TX, dumb RX” architecture is particularly relevant for biomedical applications.

C. Limitations

Several limitations should be noted: (i) the analytical channel, while physically motivated, does not capture all the complexity of particle-based simulation (e.g., receptor binding

kinetics, finite receptor capacity); (ii) the benchmark uses 500 bits, which may not fully represent steady-state performance; (iii) the Smart TX uses the reference ρ_{ref} rather than the true instantaneous decay factor, introducing a model mismatch that may limit performance in highly mobile scenarios.

IX. CONCLUSION AND FUTURE WORK

We have presented a hybrid detection architecture for mobile molecular communication that combines ISI cancellation preprocessing with Reservoir Computing detection, achieving up to 100% accuracy with only 401 trainable parameters. We further introduced a Smart Transmitter mechanism that proactively adjusts emission intensity, providing up to +4 pp accuracy improvement for ISI-sensitive detectors without requiring receiver modifications.

Our preliminary results validate three key claims: (1) physics-informed preprocessing improves ML-based detection at moderate ISI levels; (2) proactive ISI management at the transmitter is complementary to reactive detection at the receiver; and (3) Reservoir Computing provides a uniquely favorable complexity–performance trade-off for resource-constrained nanoscale devices.

Future work will focus on: (i) validation against particle-based Monte Carlo simulations for improved physical fidelity; (ii) extending the Smart TX to use adaptive γ tuning; (iii) investigating joint optimization of transmitter and receiver intelligence; and (iv) exploring the impact of mucin-like environments with enzymatic degradation and advective flow on ISI characteristics and detector performance.

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